

- The table below outlines the scripts used to complete the analysis of, “Debris flows suppressed riverine metabolism following high-severity wildfire on the Klamath River, California”.
- With the exception of the 6 days immediately before, during, and after the debris flow, high frequency data is excluded from public data repos by request of the Karuk Tribe. This high frequency data used to model metabolism and to calculate daily water quality stats (turbidity, dissolved oxygen) can be requested directly from the Karuk Tribe.
- Raw data used in the Depth_model is from Bradley 2021 and can be requested from the author.
- Scripts were checked and last run on **24 Feb 2024**, with the exception of the metabolism models, which take too long to run to justify a casual re-check.
- For questions about the data or scripts, contact Laurel Genzoli; laurel.genzoli@gmail.com

Script Name	Description	Relies on	Data out
SV-MetabCalcs.R	Runs streamMetabolizer at Seiad Valley from 2018-2023; first run is done using the Bayesian framework where K600 is estimated as a free parameter in the model; the second run is done using fixed K600 values estimated based on discharge. This scripts works iteratively with the Final_Metab_ModelOut.qmd script.	SV_clean-23072025.csv* SV_clean-MLE23072025.csv* <i>*Raw data available from Karuk Tribe on request</i>	Metab_Output/Binned/SV_GPP_ER_BIN.csv Metab_Output/Binned/SV_RDA_BIN.rda Metab_Output/Binned/SV_fits_BIN.csv Metab_Output/MLE/SV_GPP_ER_BIN.csv Metab_Output/MLE/SV_RDA_BIN.rda Metab_Output/MLE/SV_fits_BIN.csv
Depth_model.qmd	Calculate slope and intercept needed to estimate reach depth based on daily discharge at Seiad based on depth estimates from the Bradley Model.	LongProfiles/DenseMesh* LongProfiles/LongProfiles* <i>*Data available from Bradley (2021)</i>	Model coefficients to estimate reach depth based on daily flow
Final_Metab_ModelOut.qmd	This script prepares the final metabolism data frame; examines output from the binned model, builds the K600-discharge relationships for the MLE Run, then examines the MLE output for model fit/ realistic values. Builds final metab data frame and the high frequency data frame. Includes reach length calculations.	SV_clean-23072025.csv* SV_turbidity_update_2025121700173883.csv* SV_2018-2024_AllParams.csv* Metab_Output/Binned/SV_GPP_ER_BIN.csv Metab_Output/Binned/SV_fits_BIN.csv Metab_Output/Binned/SV_RDA_BIN.rda Metab_Output/MLE/SV_fits_BIN.csv Metab_Output/MLE/SV_GPP_ER_BIN.csv Metab_Output/MLE/SV_RDA_BIN.rda SV_light_2018-2023-20250723.csv <i>*Raw data available from Karuk Tribe on request</i>	met_mle_final.csv sub-daily-FIRE.csv
Final_models.qmd	This script runs the time series model and the breakpoint model. This script must be run in order for figure 2 to run in the figures script.	met_mle_final.csv	Model coefficients for final tables/figures. This has to be run in the same session for the model output that figure 2 relies on in Final_Figures.qmd
Final_Figures.qmd	File produces the final figures, including most of the SI figures. Figure 1 and Figure S6 require post processing layout.	met_mle_final.csv sub-daily-FIRE.csv	Figures!